Benson Jiang

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Education

Tufts University Expected May 2027

B.S. Computer Science and Mathematics

Relevant Coursework: Data Structures & Algorithms, Machine Structure and Assembly, Big Data, Reinforcement Learning, Machine Learning, Databases, Computer Vision, OS, Networks

Technical Skills

Programming Languages: Python, C++, C, Typescript, Javascript
Frameworks & Tools: Kubernetes, Helm, Jenkins, Flask, Git/GitHub, Linux, PyTorch, Grafana, Docker, Next.js
Databases & Cloud: PostgreSQL, Supabase, Spark, Hadoop

Experience

Software Engineering Intern May 2026 – Aug 2026

Bloomberg NYC, NY

- Designed and built a novel **CI/CD governance system** for Grafana-as-code deployments, automatically generating PRs, assigning reviewers based on internal group permissions, and validating dashboard configs with Jenkins before merge - **scaled to 100+ teams**
- Built a **rules-based metadata labeling system** that classifies Grafana dashboards by usage signals, powering a **company-wide internal search tool** used by all engineers to discover and retrieve relevant dashboards

Software Engineering Intern Aug 2025 - May 2026

Lexington Zero Waste Medford, MA

- Developed backend functionality for Lexington Zero Waste's LexShareWare platform using **Next.js, PostgreSQL, and Prisma**, supporting database access, item retrieval, and inventory management workflows
- Implemented **inventory search and retrieval** features and contributed to **admin workflows** for managing inventory and user activity across the platform

AI/ML Software Developer Aug 2024 – May 2026

MuLIP Laboratory Medford, MA

- Developed an end-to-end pipeline connecting **fNIRS neural activity to reinforcement learning agent performance**, analyzing neural responses across **12 participants** to identify brain regions associated with implicit and explicit decision-making
- Implemented **Deep Q-Learning with prioritized experience replay** to generate controlled RL policies ranging from suboptimal to $\geq 95\%$ **optimal**, enabling systematic analysis of neural responses across varying agent performance

Projects

Distributed Market Data Pipeline | C++/Python, Kafka, Redis/PostgreSQL 2026

- Built a concurrent ingestion engine that processes simulated market events from multiple feeds and normalizes them into a unified event stream
- Implemented an in-memory cache and persistent storage layer for low-latency historical and real-time queries

GPTgreen | Python, Flask, JavaScript, Next.js 2025

- Developed a full-stack app that analyzes ChatGPT usage data to estimate energy, water, and CO_2 consumption
- Built file upload and processing workflows using Flask to parse user data and generate long-term environmental impact analytics
- Won the Sustainability Track at JumboHack 2025

Publications

Mapping fNIRS Signals to Agent Performance | AAAI 2026

A Step Towards Reinforcement Learning from Neural Feedback | RLDM 2025, Extended Abstract